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06/04/2026

CS-499

Mod 5

My artifact for this milestone is my CS-340 dashboard project, originally created in the Programming Concepts course. The dashboard connects to a MongoDB database through a custom CRUD class and retrieves animal shelter records for display in a table, pie chart, and map. I selected this artifact because it demonstrates real database interaction, including reading data, filtering it with queries, and returning results to a live application.

For this milestone, I enhanced the database component by improving the reliability and security of the CRUD operations. I added better error handling, validated query inputs, and updated the read function to safely handle empty or malformed results. I also improved the filtering logic so the dashboard only requests the fields it needs, which reduces unnecessary data transfer and makes the application more efficient. These changes strengthen the database layer and make the overall system more stable.

This enhancement aligns with the program outcome related to using well-founded techniques and tools in computing practices. By improving the database logic and making the CRUD operations more robust, I met the outcome I planned for in Module One. I did not need to change my outcome-coverage plan.

While working on this enhancement, I learned how important it is to validate data coming from a database and to handle cases where queries return no results. One challenge was making sure the dashboard didn't break when the database returned an empty set or when a query didn't match any records. Adding defensive logic and improving the CRUD class helped me understand how backend reliability affects the entire application. Overall, this enhancement strengthened both the database functionality and my understanding of secure, dependable data access.